

Prediabetes and type 2 diabetes in youth: an emerging epidemic disease?



PURPOSE OF REVIEW: The prevalence of obesity in youth is increasing alarmingly among children and adolescents in the United States. The problem falls disproportionately on African-American and Hispanic children. Many of the metabolic and cardiovascular complications associated with obesity are already present during childhood and are closely linked to the concomitant insulin resistance/hyperinsulinemia and degree of obesity. Moreover, these co-morbidities persist into adulthood. RECENT FINDINGS: The progression from normal glucose tolerance to type 2 diabetes mellitus involves an intermediate stage known as prediabetes or impaired glucose regulation. Prediabetes is characterized by peripheral insulin-resistance and impaired glucose sensitivity of first-phase insulin secretion. On the other hand, in overt type 2 diabetes mellitus beta-cell failure becomes fully manifested. Progression from prediabetes to type 2 diabetes mellitus in youth is characterized by marked weight gain and further reduction in insulin secretion and insulin resistance. SUMMARY: Reverting obesity through lifestyle modification, that involves nutrition education, behavior modification and exercise, is an important step to prevent the progression to diabetes.